Review 2 Commitment

1. Project Information

Project code	FA26SE036
Project name	Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)
Team	OptiPack AI – Nguyễn Phương Mỹ Thuận, Lê Đức Trung Thi, Huỳnh Quốc Việt, Phan Huỳnh Hải Phượng
Supervisor	Thân Thị Ngọc Vân
Review 2 date	26/10/2026 (Monday), Slot 3 (12:30–14:45), room NVH.416, council HD8231
Reviewer 1	Nguyễn Thị Cẩm Hương
Reviewer 2	Nguyễn Thế Hoàng
Version	1.0 (completed after Review 2 on 08/11/2026)
Date	08/11/2026

2. Final Registration Form

Final version	1.1 (25/10/2026); changes from 1.0 in Form BM03
Approved at Review 2 (Yes/No)	
Attached file name	SWP490_FA26SE036_R2_Registration.pdf

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	Authentication and role-based access	Staff (5 roles)	5 failed logins lock the account for 15 min; a role cannot open screens or APIs outside its permissions (403)	WF-01 (used by every workflow)	Thuận
F-02	Marketplace connection and order sync	Admin, System Scheduler	A new Lazada order appears within the shop sync interval + 1 min (default ≤ 6 min); a duplicate event creates no second order	WF-01	Thuận
F-03	Order consolidation and groups	Store Owner, Shipping Coordinator	Orders with the same recipient key join one open group; a merge after a scan is rejected	WF-01	Thuận
F-04	Stock reservation and FIFO lots	Warehouse Staff	Picking consumes the oldest lot first; bin stock never becomes negative	WF-02, WF-06	Thi
F-05	Picking with missing-item handling	Warehouse Staff, Packaging Staff	A scan over the ordered quantity is rejected; a reported missing item stops the group until a decision	WF-02	Phượng
F-06	3D packing plan, approval and packing	Packaging Staff	Every item is placed or the plan cannot be approved; a parcel outside weight tolerance is held for review by another person	WF-03	Việt
F-07	Shipping with GHN	Shipping Coordinator	One GHN order per parcel; a GHN return (or, without carrier API, a second failed attempt) moves the shipment back to the warehouse	WF-04	Thuận

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-08	Returns and exchanges	Customer, Store Owner, Warehouse Staff	A request after 15 days is refused; the creator of a ticket cannot approve it; an exchange creates a replacement order	WF-05	Thi
F-09	AURELLE web store and order tracking	Customer	A placed order shows a 7-step progress bar that follows warehouse and carrier events within 1 minute	WF-05	Thi
F-10	Documents and labels	Warehouse, Packaging Staff, Shipping Coordinator	Packing slip, shipping label, picking list, manifest and bin labels download as PDF	WF-03, WF-04	Việt

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-1	Performance	Time from Lazada order creation to appearing in OptiPack	≤ sync interval + 1 min (default ≤ 6 min)	Create a test order and compare with the sync job log (TC-NFR1, manual)	Thuận
NFR-2	Performance	Packing plan computation time per order	BRKGA p95 ≤ 1 s; CP-SAT proof ≤ 6 s	be/scripts/solver-benchmark.ts and cpsat-benchmark.ts on 300 orders (TC-NFR2)	Thi
NFR-3	Security	Failed logins before lockout; role checks	Lock after 5 failures for 15 min; 100% of staff endpoints return 403 to an unauthorized role	Automated auth / role-guard tests (TC-NFR3, TC-F01-02, TC-F01-03)	Thuận
NFR-4	Reliability	Status write-back success	Up to 8 retries with back-off; 0 lost changes	Outbox test: AURELLE returns errors, job retries until the limit (TC-NFR4)	Thuận
NFR-5	Data integrity	Stock correctness under repeated / concurrent scans	0 negative bin stock; a repeated scan with the same client event id is not counted twice	Send the same pick scan twice with the same client_event_id (TC-NFR5)	Phượng
NFR-6	Availability	Packing without the optimization service	Plan still produced when the CP-SAT service is down (label heuristic)	Stop the packer service and compute a plan (TC-NFR6)	Thi

5. Out of Scope

Item	Reason
Order sync for Tiki, TikTok Shop, Shopee, Facebook Marketplace	Partner API access constraints (BM03 #1). Tiki: the team is registering a tax code for a Tiki seller account (stated at Review 2); the Tiki adapter added on 31/10/2026 is outside the Review 2 demo
Write-back of order status to Lazada	Lazada is read-only by design; only AURELLE receives status and stock write-back

Item	Reason
Live carriers other than GHN	GHN (staging) is the only carrier with a live API; others use sample rate tables
Excel / PDF analytical report export	Only CSV export of the action history (BM03 #2)
Mobile app (Flutter)	Planned for Iteration 3 (Design Brief G1); scanning is done on the web portal

6. Workflow Commitments

6.1 Workflow WF-01

Workflow name	WF-01 Order Intake & Consolidation
Owner (one member)	Nguyễn Phương Mỹ Thuận
Supporting members	Phan Huỳnh Hải Phượng
Actors	Customer, Marketplace Platform (Lazada / AURELLE), OptiPack System, Store Owner / Shipping Coordinator, Warehouse Staff
Trigger	Customer places an order on Lazada or AURELLE
Business outcome	Order imported, grouped by recipient, stock reserved, picker assigned
Main entity state path	Order Group: awaiting_packaging → picking (or canceled)
Activity diagram (Report 4 section)	Report 4 section 3.2 (Figure 6)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	Authentication and role-based access	5 failed logins lock the account for 15 min; a role cannot open screens or APIs outside its permissions (403)	TC-F01-01 … 03
F-02	Marketplace connection and order sync	A new Lazada order appears within the shop sync interval + 1 min (default ≤ 6 min); a duplicate event creates no second order	TC-F02-01 … 03
F-03	Order consolidation and groups	Orders with the same recipient key join one open group; a merge after a scan is rejected	TC-F03-01 … 03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	*Main*	*Lazada test shop connected; order for 2 SKUs in stock*	*Import → group → reserve → assign picker*	*Group in picking, picker assigned*	*Thuận*
DS-02	*Exception*	*Order already cancelled when imported*	*Same flow with the cancelled order*	*Group cancelled automatically, reservation released; nothing is picked*	*Thuận*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*9*	*9*	*Thuận*
End-to-end workflow test (≥ 1)	*1 (TC-WF01)*	*0 (manual)*	*Thuận*
State transition tests (≥ 1 per transition)	*2 (TC-ST01, TC-ST03)*	*2*	*Thuận*
Total for this workflow	*12*	*11*	*Thuận*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*

Criterion	Committed
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

6.2 Workflow WF-02

Workflow name	WF-02 Picking
Owner (one member)	Phan Huỳnh Hải Phượng
Supporting members	Huỳnh Quốc Việt
Actors	Warehouse Staff, OptiPack System, Packaging Staff
Trigger	Group in picking with an assigned picker
Business outcome	All items picked, or shortage decided (ship available / re-pick)
Main entity state path	Order Group: picking → picked, or → partial_needs_review → picked / picking
Activity diagram (Report 4 section)	Report 4 section 3.3 (Figure 7)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-05	Picking with missing-item handling	A scan over the ordered quantity is rejected; a reported missing item stops the group until a decision	TC-F05-01 … 03
F-04	Stock reservation and FIFO lots	Picking consumes the oldest lot first; bin stock never becomes negative	tests counted in WF-06

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03	*Main*	*Group in picking with stock in bins*	*Scan all items bin by bin → confirm*	*Group picked; oldest lots consumed*	*Phượng*
DS-04	*Exception*	*SKU not found in its bin*	*Report missing item*	*Group partial_needs_review; Store Owner and Packaging Staff notified*	*Phượng*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*3*	*3*	*Phượng*
End-to-end workflow test (≥ 1)	*1 (TC-WF02)*	*0 (manual)*	*Phượng*
State transition tests (≥ 1 per transition)	*8 (TC-ST02, 04, 05, 06, 10, 11, 12, 13)*	*5*	*Phượng*
Total for this workflow	*12*	*8*	*Phượng*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

6.3 Workflow WF-03

Workflow name	WF-03 Packing
Owner (one member)	Huỳnh Quốc Việt
Supporting members	Lê Đức Trung Thi
Actors	OptiPack System, Packaging Staff, Reviewer (other staff / Admin)
Trigger	Group reaches picked
Business outcome	Every parcel sealed and weighed; boxes and materials deducted
Main entity state path	Packing Plan: computing → ready → approved → packing → packed (or rejected / failed / superseded); Order Group: picked → pending_approval → approved_for_packing → packed
Activity diagram (Report 4 section)	Report 4 section 3.4 (Figure 8)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	3D packing plan, approval and packing	Every item is placed or the plan cannot be approved; a parcel outside weight tolerance is held for review by another person	TC-F06-01 … 03
F-10	Documents and labels	Packing slip, shipping label, picking list, manifest and bin labels download as PDF	TC-F10-01 … 03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05	*Main*	*Picked group, SKU profiles confirmed*	*Compute plan → approve → scan → seal → weigh*	*Group packed*	*Việt*
DS-06	*Exception*	*Parcel weight outside tolerance*	*Seal and weigh with a wrong weight*	*Parcel held and reviewed by another person*	*Việt*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*6*	*6*	*Việt*
End-to-end workflow test (≥ 1)	*1 (TC-WF03)*	*0 (manual)*	*Việt*
State transition tests (≥ 1 per transition)	*13 (TC-ST07, 08, 09, 14–21, 23, 24)*	*10*	*Việt*
Total for this workflow	*20*	*16*	*Việt*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

6.4 Workflow WF-04

Workflow name	WF-04 Shipping (GHN and Express Self-Delivery)
Owner (one member)	Nguyễn Phương Mỹ Thuận
Supporting members	Huỳnh Quốc Việt
Actors	Shipping Coordinator, OptiPack System, GHN (Carrier), Customer, Warehouse Staff
Trigger	Shipping Coordinator creates a shipment for a packed group
Business outcome	Parcels delivered, or returned to the warehouse with a failed-delivery ticket
Main entity state path	Shipment: awaiting_pickup → in_transit → out_for_delivery → delivered / delivery_failed → returning_to_warehouse → returned_to_warehouse (or cancelled before pickup)
Activity diagram (Report 4 section)	Report 4 section 3.5 (Figure 9)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-07	Shipping with GHN	One GHN order per parcel; a GHN return (or, without carrier API, a second failed attempt) moves the shipment back to the warehouse	TC-F07-01 … 03
F-10	Documents and labels	Packing slip, shipping label, picking list, manifest and bin labels download as PDF	tests counted in WF-03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-07	*Main*	*Packed group; GHN staging configured*	*Quote → create shipment → GHN pickup → delivered*	*Group delivered; AURELLE customer sees every step*	*Thuận*
DS-08	*Exception*	*GHN reports failed delivery and return (manual status buttons in the demo)*	*Fail → return → receive at warehouse*	*Shipment returned_to_warehouse ; failed-delivery return ticket created*	*Thuận*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*3*	*3*	*Thuận*
End-to-end workflow test (≥ 1)	*1 (TC-WF04)*	*0 (manual)*	*Thuận*
State transition tests (≥ 1 per transition)	*4 (TC-ST22, 25,*	*4*	*Thuận*

Test type	Count	Automated in CI	Responsible member(s)
	26, 27)		
Total for this workflow	*8*	*7*	*Thuận*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

6.5 Workflow WF-05

Workflow name	WF-05 Return / Exchange via AURELLE
Owner (one member)	Lê Đức Trung Thi
Supporting members	Huỳnh Quốc Việt
Actors	Customer, AURELLE Website, OptiPack System, Store Owner, Warehouse Staff
Trigger	Customer submits a return / exchange request on AURELLE
Business outcome	Items inspected; refund recorded or replacement order created
Main entity state path	Return Request: requested → awaiting_receipt → received → closed (or rejected / canceled)
Activity diagram (Report 4 section)	Report 4 section 3.6 (Figure 10)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-08	Returns and exchanges	A request after 15 days is refused; the creator of a ticket cannot approve it; an exchange creates a replacement order	TC-F08-01 … 03
F-09	AURELLE web store and order tracking	A placed order shows a 7-step progress bar that follows warehouse and carrier events within 1 minute	TC-F09-01 … 03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-09	*Main*	*Delivered AURELLE order within 15 days*	*Request → approve → receive → inspect*	*Ticket closed; refund or replacement order*	*Thi*
DS-10	*Exception*	*Request more than 15 days after delivery*	*Submit the request*	*Request refused with reason*	*Thi*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*6*	*6*	*Thi*

Test type	Count	Automated in CI	Responsible member(s)
End-to-end workflow test (≥ 1)	*1 (TC-WF05)*	*0 (manual)*	*Thi*
State transition tests (≥ 1 per transition)	*6 (TC-ST28 … TC-ST33)*	*6*	*Thi*
Total for this workflow	*13*	*12*	*Thi*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

6.6 Workflow WF-06

Workflow name	WF-06 Stock Receiving by Lot
Owner (one member)	Lê Đức Trung Thi
Supporting members	Phan Huỳnh Hải Phượng
Actors	Warehouse Staff, OptiPack System, AURELLE Website
Trigger	Goods arrive from the supplier
Business outcome	Stock received by lot; shortage groups re-checked; available stock pushed to AURELLE
Main entity state path	Stock Lot created; stock-shortage flag of groups cleared
Activity diagram (Report 4 section)	Report 4 section 3.7 (Figure 11)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-04	Stock reservation and FIFO lots	Picking consumes the oldest lot first; bin stock never becomes negative	TC-F04-01 … 03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-11	*Main*	*Warehouse with an empty bin*	*Assign SKU to bin → create receipt*	*Lot created; stock increased*	*Thi*
DS-12	*Exception*	*Receipt exceeds bin capacity*	*Create the receipt*	*Receipt rejected with over-capacity detail*	*Thi*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*3*	*3*	*Thi*
End-to-end workflow test (≥ 1)	*1 (TC-WF06)*	*0 (manual)*	*Thi*
State transition tests (≥ 1 per transition)	*0 (—)*	*0*	*Thi*

Test type	Count	Automated in CI	Responsible member(s)
Total for this workflow	*4*	*3*	*Thi*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	*1 of 1 (100%)*
Test pass rate for this workflow	*100% (≥ 90% required)*
Open Critical / High defects	*0*

Reviewer note

Note	

7. Individual Commitments

7.1 Member 1

Full name	Nguyễn Phương Mỹ Thuận
Student ID	SE171113
Role in team	Leader, back-end developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-01, WF-04*	
Mandatory features owned	*F-01, F-02, F-03, F-07*	
Modules / entities owned	*auth, users, marketplace-integration, marketplace-webhooks, orders, order-groups (consolidation), shipping, carriers, shipments, marketplace-writeback*	
Design decisions owned (D-IDs)	*D1, D3, D4, D5*	
NFRs responsible	*NFR-1, NFR-3, NFR-4*	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*23 (WF-01 12, WF-04 8, TC-NFR1, 3, 4)*	*Defense 1.1*
Of which automated in CI	*20*	*Defense 1.1*
Demo scenarios presented	*DS-01, DS-02, DS-07, DS-08*	*Defense 1.1*
Required fixes owned (section 9)	*RF-06, RF-11*	*see section 9*
Documentation sections (Report / section)	*Reports 1–5, Design Brief, BM04; SDD 3.1–3.7, 4.1, 4.2, 4.6, 4.8*	*15/11/2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	

7.2 Member 2

Full name	Lê Đức Trung Thi
Student ID	DE180553
Role in team	Back-end developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-05, WF-06*	
Mandatory features owned	*F-04, F-08, F-09*	

Type	Items (IDs / names)	Notes
Modules / entities owned	*packing solver (BRKGA, CP-SAT packer), returns, warehouse stock receipts and lots*	
Design decisions owned (D-IDs)	*D2*	
NFRs responsible	*NFR-2, NFR-6*	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*19 (WF-05 13, WF-06 4, TC-NFR2, 6)*	*Defense 1.1*
Of which automated in CI	*16*	*Defense 1.1*
Demo scenarios presented	*DS-09, DS-10, DS-11, DS-12*	*Defense 1.1*
Required fixes owned (section 9)	*RF-01, RF-02, RF-05, RF-09, RF-12*	*see section 9*
Documentation sections (Report / section)	*SDD 1.1, 3.8, 4.3, 4.7*	*15/11/2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	

7.3 Member 3

Full name	Huỳnh Quốc Việt
Student ID	SE182482
Role in team	Front-end developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-03*	
Mandatory features owned	*F-06, F-10*	
Modules / entities owned	*Packing Station UI (3D viewer, plan review, packing session); picking, shipping and returns screens*	
Design decisions owned (D-IDs)	—	
NFRs responsible	—	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*20 (WF-03)*	*Defense 1.1*
Of which automated in CI	*16*	*Defense 1.1*
Demo scenarios presented	*DS-05, DS-06*	*Defense 1.1*
Required fixes owned (section 9)	*RF-03, RF-04, RF-07, RF-10, RF-13*	*see section 9*
Documentation sections (Report / section)	*SDD section 2, SDD 4.5, 4.9*	*15/11/2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	

7.4 Member 4

Full name	Phan Huỳnh Hải Phượng
Student ID	SE192623
Role in team	Front-end developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-02*	
Mandatory features owned	*F-05*	
Modules / entities owned	*Admin portal (accounts, roles, leave); admin warehouse and SKU screens; marketplace connection screens*	
Design decisions owned (D-IDs)	*—*	
NFRs responsible	*NFR-5*	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*13 (WF-02 12, TC-NFR5)*	*Defense 1.1*
Of which automated in CI	*9*	*Defense 1.1*
Demo scenarios presented	*DS-03, DS-04*	*Defense 1.1*
Required fixes owned (section 9)	*RF-08, RF-14*	*see section 9*
Documentation sections (Report / section)	*SRS 2.2, 3.4.3, SDD 4.4, 4.10*	*15/11/2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	*Thuận*	*F-01, F-02, F-03*	*1 / 1*	*12*	*11*
WF-02	*Phượng*	*F-05 (F-04)*	*1 / 1*	*12*	*8*
WF-03	*Việt*	*F-06, F-10*	*1 / 1*	*20*	*16*
WF-04	*Thuận*	*F-07 (F-10)*	*1 / 1*	*8*	*7*
WF-05	*Thi*	*F-08, F-09*	*1 / 1*	*13*	*12*
WF-06	*Thi*	*F-04*	*1 / 1*	*4*	*3*
Total	*4 members*	*10 features*	*6 / 6*	*69 (+6 NFR = 75)*	*57 (+4 NFR = 61)*

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Thuận	*WF-01, WF-04*	*F-01, F-02, F-03, F-07*	*23*	*20*	*4*	
Thi	*WF-05, WF-06*	*F-04, F-08, F-09*	*19*	*16*	*4*	
Việt	*WF-03*	*F-06, F-10*	*20*	*16*	*2*	
Phượng	*WF-02*	*F-05*	*13*	*9*	*2*	
Total	*6*	*10*	*75*	*61*	*12*	

8.3 Committed Numbers

Number of main workflows	6
Number of demo scenarios (main / exception)	6 / 6
Committed functional test cases	75
Of which automated in CI	61 (81%)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	*100% run end-to-end*	*100% run end-to-end*
Demo scenarios	*100% main; ≥ 80% exception*	*100% main; ≥ 80% exception*
Test case count	*≥ committed number*	*≥ committed number*
Test pass rate	*≥ 90% overall; 100% main-path tests of mandatory features*	*≥ 90% overall; 100% main-path tests of mandatory features*
Automation	*≥ 50% run in CI*	*≥ 50% run in CI*
Defects	*No open Critical or High defects*	*No open Critical or High defects*
Individual	*Each member demos the workflow they own and meets their delivery commitments*	*Each member demos the workflow they own and meets their delivery commitments*

9. Required Fixes

ID	Item	Related workflow	Owner	Due
RF-01	State machines of Packing Plan, Shipment, Return Request in draw.io; redraw Order Group with all 33 transitions (BM02 F03, F12)	WF-03, WF-04, WF-05	Thi	15/11/2026
RF-02	Add Google OAuth and SMTP to the architecture diagram (BM02 F04)	All	Thi	15/11/2026
RF-03	Extend the logical ERD to all main entities and collections (BM02 F05, F23)	All	Việt	15/11/2026
RF-04	Database script with the collections and indexes of the ERD (BM02 F06)	All	Việt	15/11/2026
RF-05	Automated end-to-end tests pick → pack → ship → return (BM02 F14, Design Brief G4)	WF-02 … WF-05	Thi	04/12/2026
RF-06	Package diagram and class / sequence diagrams of SDD section 4 (BM02 F16), each section by its owner	All	Thuận	15/11/2026
RF-07	Mobile app (Flutter) for picking / packing scan (BM02 F01, Design Brief G1)	WF-02, WF-03	Việt	20/11/2026
RF-08	Packaging profile screenshot in SRS 3.4.3 (BM02 F20)	WF-03	Phượng	15/11/2026
RF-09	Redraw the architecture diagram by hand in draw.io (Review 2 minutes). Done: SDD Figure 1, commit e2f505c	All	Thi	03/11/2026
RF-10	ERD: correct the inheritance notation and shapes (Review 2 minutes). Done: SDD Figure 3, commit e2f505c	All	Việt	06/11/2026
RF-11	Activity diagrams: split the warehouse flow into supplier receiving and customer orders; redraw by hand in draw.io (Review 2 minutes). Done: SDD Figures 4–11, commit e2f505c	All	Thuận	07/11/2026
RF-12	Define "optimize" with measurable metrics: run a smallest-fitting-box baseline and the solver on the same orders and compare total cost, fill ratio and cartons per order; state the order-to-packed time saved by consolidation (Review 1 and Review 2 minutes)	WF-01, WF-03	Thi	15/11/2026
RF-13	Show the packing rules applied to each order (company box-selection rules and the reason for the chosen box) next to the packing plan (Review 2 minutes)	WF-03	Việt	04/12/2026
RF-14	Consolidation screen: merge and split groups and show which channel and orders each group came from (Review 2 minutes)	WF-01	Phượng	04/12/2026

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1	*Nguyễn Thị Cẩm Hương*		
Reviewer 2	*Nguyễn Thế Hoàng*		
Supervisor	*Thân Thị Ngọc Vân*		
Team leader	*Nguyễn Phương Mỹ Thuận*		